Checking the answer in (4), (5) & (6) by substituting back into the system:

$$\frac{dx}{dt} = \frac{d}{dt}\left(e^{-t}\cdot C_1\right) = -e^{-t}C_1 \equiv -x \checkmark$$

$$\frac{dy}{dt} = \frac{d}{dt}\left[e^{t}\cdot C_2\right] = e^{t}\cdot C_2 \equiv y \checkmark$$

multiply top by $\boxed{1 = e^{-t}\cdot e^{t}}$

$$\frac{dz}{dt} = \frac{d}{dt}\left[\tan(C_1C_2t+C_3)\right] = \frac{C_1\cdot C_2}{\cos^2(C_1C_2t+C_3)} = \underbrace{C_1e^{-t}}_{x}\cdot\underbrace{C_2e^{t}}_{y}\cdot\Big[1+\underbrace{\tan^2(C_1C_2t+C_3)}_{z^2}\Big] = xy(z^2+1).$$

$\boxed{1+\tan^2\alpha = \frac{1}{\cos^2\alpha}}$ apply bottom

$\checkmark$ ▨

Problem 3

(a) (i) Find the general sol'n of the system by eliminating one var'le:

$$\begin{cases} \dfrac{dx_1}{dt} = x_2 & (1) \\[2ex] \dfrac{dx_2}{dt} = 8 & (2) \end{cases}$$

(ii) Show by substituting back that your sol'n is correct.

Sol'n:

(i) by (2), using sep'n of var's:

$$\int dx_2 = \int 8\,dt + C_1 \Rightarrow$$

(3) $$\boxed{x_2(t) = 8t + C_1}$$

Now plugging this (3) into (1) eliminates x_2 in eq'n (1):

$$\frac{dx_1}{dt} = 8t + C_1$$ applying sep'n var's to it

$$\int dx_1 = \int (8t+C_1)\,dt + C_2 \Rightarrow$$

(4) $$\boxed{x_1(t) = 4t^2 + C_1t + C_2}$$

(ii) Checking the pair of sol'ns in (3) & (4) by back sub-n:

By (1), $\dfrac{dx_1}{dt} = \dfrac{d}{dt}\left(4t^2+C_1t+C_2\right) = 8t + C_1 \equiv x_2$ ✓

By (2), $dx_2/dt = \dfrac{d}{dt}(8t+C_1) \equiv 8$ ✓, thus verified. ▨

(b) Find the sol'n for (1)+(2) using the IC's $x_1(0)=5$, $x_2(0)=-3$:

Sol'n: By (4), $x_1(0) = 4(0)^2 + C_1\cdot 0 + C_2 = 5 \Rightarrow \boxed{C_2 = 5}, \Rightarrow \boxed{x_1(t) = 4t^2 - 3t + 5}$

By (3), $x_2(0) = 8\cdot 0 + C_1 = -3 \Rightarrow \boxed{C_1 = -3} \Rightarrow \boxed{x_2(t) = 8t - 3}$. ▨

Problem 4. Find the general sol'n of the system of eq'ns

(1) .. $\frac{dx}{dt} = -x$

(2) .. $\frac{dy}{dt} = y$

(3) ... $\frac{dz}{dt} = xy(z^2+1)$

Hint: How many arbitrary constants should the sol'n have? Check your answer by substituting back to (1), (2), (3).

Sol'n:

Using the sep'n of variables for (1) & (2):

$$\frac{dx}{dt} = -x \Rightarrow \int \frac{dx}{x} = \int -dt + \ln(c_1) \Rightarrow \ln|x| = \ln(e^{-t} \cdot c_1)$$

for $c_1 > 0$

(4) $\boxed{x = e^{-t} \cdot c_1}$, and similarly,

$$\frac{dy}{dt} = y \Rightarrow \int \frac{dy}{y} = \int dt + \ln(c_2) \Rightarrow \ln|y| = \ln(e^{t} \cdot c_2)$$

for some $c_2 > 0$

(5) $\boxed{y = e^{t} \cdot c_2}$

And finally substituting (4) & (5) into (3) gives

$$\frac{dz}{dt} = e^{-t} \cdot c_1 \cdot e^{t} \cdot c_2 \, (z^2+1)$$

$$\frac{dz}{dt} = c_1 \cdot c_2 \, (z^2+1)$$, applying sep'n of var's:

$$\int \frac{dz}{z^2+1} = \int c_1 \cdot c_2 \, dt + c_3$$

(6) $\boxed{z(t) = \tan(c_1 c_2 t + c_3)}$ $\Rightarrow$

Checking (4), (5), (6) by subs'ing into the system:

$\frac{dx}{dt} = \frac{d}{dt}(e^{-t} \cdot c_1) = -e^{-t} \cdot c_1 \equiv -x$ ✓

$dy/dt = \frac{d}{dt}(e^{t} \cdot c_2) = e^{t} \cdot c_2 \equiv y$ ✓ algebraic trick

$\frac{dz}{dt} = \frac{d}{dt}(\tan(c_1 c_2 t + c_3)) = \frac{c_1 c_2 \cdot [e^{-t} \cdot e^{t}]}{\cos^2(c_1 c_2 t + c_3)} = xy(z^2+1)$ ✓